

Evaluating Retrieval-Augmented Generation Systems

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Abstract

Evaluating retrieval-augmented generation (RAG) requires more than attaching an answer-quality score to a pipeline whose retrieval, context construction, and generation stages can fail independently [7, 17, 20]. This paper organizes the problem around three questions: what outcome is being measured, whether the protocol measures that outcome validly, and whether the resulting conclusion survives changes in queries, corpora, domains, and deployment conditions [6, 12, 13]. The evidence base consists of 320 retrieved records and a 48-paper core set, interpreted conservatively because most of the available evidence is abstract-level rather than full-text [18, 11, 10]. Results reported across this literature show that retrieval relevance, evidence coverage, answer faithfulness, source attribution, and user utility are related but non-interchangeable targets [14, 16, 2]. They also show that evaluator rankings can reverse across datasets, while failures that matter in deployment can escape otherwise strong automated scores [7, 6]. A defensible RAG evaluation should therefore report a stage-resolved metric portfolio, human calibration for the intended domain, robustness tests under plausible shifts, and the cost and governance conditions under which the measured system will operate [4, 5, 8, 24].

1 Introduction

Retrieval-augmented generation is often evaluated as though it were a single model, although its observed answer is produced by a chain of document preparation, retrieval, context selection, and generation decisions. A correct answer can conceal weak retrieval, and a well-supported answer can still cite the wrong entity or fail the user’s task [13, 6]. Conversely, a retriever can rank relevant passages highly without supplying enough evidence for a faithful answer [20, 2]. End-to-end accuracy alone therefore cannot identify which part of the system improved or which failure remains.

The evaluation target must be named before a score can be interpreted. Retrieval relevance asks whether useful items are ranked highly; evidence coverage asks whether the returned context is sufficient; faithfulness asks whether claims follow from that context; attribution asks whether claims are connected to the correct sources; and utility asks whether the answer helps with the task [7, 14, 16]. Collapsing these constructs into one number makes compensating errors look like progress and prevents a system owner from choosing the right repair.

Measurement validity creates a second layer of difficulty. Different datasets vary in query form, document type, reference assumptions, and acceptable answers, while rule-based, learned, and LLM-based judges make different errors [23, 10, 21]. Ding et al.’s cross-dataset results show that even attribution-metric rankings can invert, so success on one benchmark does not establish that the evaluator is portable [7]. Evaluation is consequently part of the system claim, not a neutral instrument applied after development.

A third concern is operational validity: the system encountered after deployment may differ from the one represented by a static benchmark. Queries become longer or noisier, corpora are updated, contradictory evidence appears, and latency or annotation budgets constrain what can be measured [21, 23, 8]. Domain transfer can also introduce failure modes that generic faithfulness checks were not designed to detect [6, 4]. Robustness and deployment evidence must therefore accompany average benchmark performance.

The paper develops these three layers using a candidate pool of 320 records and a 48-paper core set. Because the available evidence is predominantly abstract-level, the analysis treats reported findings as bounded evidence and avoids inferring unreported controls or implementation details. Sections 3 through 5 move from evaluation targets, to protocol validity, to robustness and deployment, ending in a practical minimum contract for reporting RAG evaluations [12, 5, 24].

2 Related Work

Existing RAG reviews supply useful taxonomies of architectures, applications, and evaluation resources, but those organizing schemes do not by themselves make empirical results comparable [17, 3, 24]. A method-centered review may group systems that share a retrieval architecture even when their tasks and metrics differ, whereas an application review may combine systems that face similar domains but use incompatible evidence and judging protocols. The present analysis uses prior reviews for vocabulary and scope, then organizes the empirical comparison around measurable failure locations and validity threats.

One body of work treats evaluation as pipeline diagnosis. Retrieval studies examine ranking, chunking, evidence sufficiency, and the propagation of retrieval errors into answers [20, 2]. System-level frameworks likewise separate preprocessing, retrieval, context packing, and generation so that final accuracy is not mistaken for an explanation of system behavior [13]. This line establishes why evaluation targets need stage-specific observables.

A second body of work studies grounded generation and evaluator design. ARES scores context relevance, answer faithfulness, and answer relevance, while RAGVUE is motivated by the need to decompose errors that aggregate metrics obscure [16, 12]. Attribution research further shows that evaluator performance depends on the dataset and claim form, and source-attribution work examines the cost of assigning contribution to retrieved documents [7, 14]. These studies shift attention from obtaining a score to validating what the score means.

Benchmark research makes the same point from the data side. Long narrative queries, imperfect visual queries, and medical question answering each stress different capabilities and expose different failure distributions [21, 23, 10]. AuthorityBench reports that adding webpage text can worsen authority judgments, an example of apparently richer input changing the construct being assessed rather than simply increasing information [22]. Benchmark breadth is useful only when task composition and aggregation remain visible.

Deployment-oriented studies add failures that laboratory metrics can miss. Clinical work identifies deceptive grounding and contradiction-sensitive retrieval as risks even when cited documents are real and superficially relevant [6, 8]. Industry evidence also describes systems that remain prototype-heavy and rely substantially on human evaluation, with security and data quality receiving more attention than ethics, bias, or scalability [4]. These findings motivate an explicit operational-validity layer. The contribution here is thus a joint reading of target validity, measurement validity, and operational validity. It does not claim an exhaustive history or a universal ranking of evaluators. Instead, it asks what evidence would justify a decision about a RAG system when benchmark scope, metric behavior, cost, and deployment risk are all part of that decision [5, 9, 1].

3 Evaluation Targets & Failure Decomposition

A RAG score is interpretable only after the object of evaluation has been specified. Retrieval quality, evidence sufficiency, answer faithfulness, and source attribution locate different faults in the pipeline, and improvement in one does not guarantee improvement in the others [20, 7, 6]. The two subsections separate upstream access to evidence from downstream use of that evidence. The first examines ranking and coverage; the second examines claim support, citation correctness, and answer utility, including cases where a response looks grounded while relying on the wrong source entity [11, 2, 14].

3.1 Retrieval quality and evidence coverage

A valid retrieval evaluation must report three distinct objects: the order of retrieved items, the amount of answer-bearing evidence in the returned context, and the contribution of that context to the final answer. Rank metrics can show whether relevant items appear near the top, but they do not establish that the selected passages jointly cover a multi-part question [20, 9]. Answer utility is further downstream and can be affected by generation even when retrieval is unchanged [11, 7]. Reporting all three prevents a gain at one stage from being presented as an unexplained end-to-end improvement.

Chunking illustrates why the retrieval configuration belongs in the evaluation record. Taiwo et al. report that structure-aware chunking improves top-K retrieval effectiveness while reducing computational cost relative to semantic and baseline strategies [20]. Belinchon et al. observe a related trade-off at generation time: larger chunks can supply broader information, but they also introduce noise and raise computation [2]. A fair experiment should therefore vary chunk construction and retrieval depth explicitly, measuring coverage and cost in addition to a rank score.

Input modality and corpus structure can change the same trade-off. In a multimodal setting, Sauc et al. report a 157% retrieval-quality improvement over Naive RAG with approximately 50 ms of added latency, and a small Qwen2-VL model achieved BERTScore results comparable to cloud models [18]. That result provides a useful latency-aware anchor, but it should not be read as a general ranking against text-only systems because the query and evidence modalities differ. On heterogeneous corpora and multi-step queries, Kovalskii et al. report quality degradation for flat Naive RAG pipelines that lack explicit workflows [9].

Downstream evidence makes clear why retrieval recall cannot stand alone. C-MIG is reported to outperform other RAG-RL methods on four medical benchmarks in both in-domain and out-of-domain settings, as well as general-purpose clinical LLM baselines [11]. This is evidence about task performance under a particular optimization and benchmark design, not direct proof that its retriever dominates on a shared rank metric. System comparisons need both the retrieval-stage observables and the answer-stage outcome to determine where the gain arose [11, 17].

Evaluator instability can otherwise obscure the relationship between the two stages. On AttributionBench and HAGRID, metric rankings varied substantially across datasets; for AttributedQA versus LFQA, the reported Kendall correlation was -0.64, and an NLI scorer fell from 0.90 AU-ROC on short claims to 0.53 on long-form answers while BERTScore reached 0.91 [7]. Evidence from benchmark tasks should not be generalized across retrieval quality and evidence coverage without matching success rate and budget and cost model conditions [7]. Although these are attribution results, they demonstrate a general evaluation rule: retrieval and coverage metrics should be validated on the target task rather than selected only by average performance elsewhere.

The minimum useful retrieval report should identify the query population, corpus snapshot, chunking policy, top-k and reranking settings, rank metrics, evidence-coverage measure, answer-level outcome, and latency or cost budget. A limitation of the present comparison is that several findings are available only at abstract level, leaving some controls and uncertainty estimates unclear. Perturbing at least the query form, corpus composition, and retrieval depth would reveal whether the conclusion depends on one convenient configuration [20, 9, 18].

3.2 Generation faithfulness and source attribution

Generation evaluation should distinguish whether a claim is supported by the supplied context, whether it is assigned to the correct source, whether citations are complete and precise, and whether the answer is useful. These properties can disagree: a response may paraphrase a retrieved passage faithfully yet attach it to the wrong entity, or it may cite appropriate sources while omitting evidence needed for a complete answer [14, 6]. A single groundedness score cannot reveal which of these failures occurred.

Automated frameworks already expose part of this structure. ARES evaluates context rel-

evance, answer faithfulness, and answer relevance rather than representing a RAG system with one undifferentiated measure [16]. RAGVUE is similarly motivated by the problem that existing metrics hide whether an error originated in retrieval, reasoning, or grounding [12]. These decompositions are most useful when each dimension is reported separately and tied to a defined unit of analysis, such as claims, citations, or complete answers.

Deceptive grounding demonstrates the limit of ordinary faithfulness checks. Caruzzo et al. describe a clinical response that passes automated checks, uses real citations, and contains no conventional hallucination, yet presents evidence about drug Y as though it concerned queried drug X [6]. Cross-study interpretation in generation faithfulness and source attribution remains conditional on comparable benchmark tasks, success rate, and budget and cost model assumptions [6]. Entity-correct attribution must therefore be tested independently of textual entailment and citation existence.

Source contribution introduces a different cost problem. Nematov et al. note that Shapley-style attribution for RAG remains challenging because each utility evaluation can require an expensive LLM call [14]. Context granularity also affects the result: broad chunks may increase the chance that supporting material is present while adding irrelevant text and computational expense [2]. An attribution protocol should state the context unit, utility function, sampling approximation, and number of model calls so that explanatory fidelity is not detached from evaluation cost.

The evidence supports a portfolio rather than a universal faithfulness metric. Claim-level entailment can detect unsupported statements, whereas entity checks can detect misattribution; citation precision and completeness can test documentation, whereas task-specific human judgments can assess whether a supported answer is actually adequate [16, 6, 12]. Correlations among these measures should be reported, but high agreement on familiar data should not be treated as proof that one measure substitutes for all the others.

For generation, a reproducible evaluation record should include claim segmentation rules, source-to-claim links, the handling of partially supported claims, judge prompts or model versions, human adjudication procedures, and inference cost. The open issue is how to retain this diagnostic detail at production scale without replacing costly human validation with an automated metric that is blind to the domain’s most consequential error [14, 6].

4 Evaluation Protocols & Measurement Validity

Once the target is clear, the next question is whether the protocol measures it reliably. Dataset composition, task formulation, reference construction, metric choice, and judge calibration jointly determine what a reported result can support [23, 10, 7]. The chapter first considers how benchmarks expose or suppress particular failures, then compares scalable automated judging with human validation. The connection is important because evaluator agreement is meaningful only for the populations and constructs on which that agreement was established [16, 12, 6].

4.1 Dataset, task, and metric design

A benchmark defines which users, queries, documents, and failures are visible to an evaluation. Two studies that both report answer accuracy may still test different constructs if one uses short factoid questions and the other requires long-form synthesis, multimodal search, or clinical reasoning [21, 23, 10]. Dataset composition, task instructions, references, metrics, and aggregation should therefore be treated as one protocol rather than as separable implementation details. Additional work in dataset, task, and metric design keeps returning to corpus and query provenance as a comparison pressure [15].

Stress tests are valuable because they make hidden assumptions observable. The cited result remains specific to benchmark tasks; broader comparison in dataset, task, and metric design requires explicit success rate and budget and cost model reporting [21]. Work on imperfect visual queries tests geometric distortion, quality degradation, and semantic ambiguity, conditions reported to

cause severe retrieval failures [23]. These tasks should not simply be pooled with ordinary queries; stratified results show whether the system is robust or merely benefits from an easier mixture.

Domain benchmarks expose a different form of coverage. Li et al. motivate MRAG-Bench by the absence of a comprehensive medical RAG benchmark despite rapid adoption in scientific and clinical question answering [10]. AuthorityBench reports that incorporating webpage text consistently degrades authority judgments, suggesting that authority is not reducible to stylistic signals in the page content [22]. Together, these findings show that adding domains or input fields can change the measured construct as well as the sample size.

Breadth alone does not ensure metric validity. ARES reports evaluation across eight knowledge-intensive tasks drawn from KILT, SuperGLUE, and AIS while using a few hundred human annotations for calibration [16]. By contrast, Ding et al. Cross-study interpretation in dataset, task, and metric design remains conditional on comparable benchmark tasks, success rate, and budget and cost model assumptions [7]. The first result supports scalable multi-task evaluation; the second warns that scale does not eliminate the need for target-population validation.

Reference and aggregation choices should receive the same scrutiny. A metric may reward overlap with one accepted answer even when several answers are valid, and an average can hide a catastrophic failure concentrated in one query type. Reporting per-stratum distributions, uncertainty, missing cases, and disagreements among metrics is more informative than publishing only a leaderboard mean [7, 23]. Releasing query construction and scoring procedures is also necessary for reproducibility and leakage analysis.

A credible protocol card would state the target population, corpus date, sampling process, task instructions, reference provenance, unit of scoring, metric implementation, aggregation rule, uncertainty estimate, and evaluation cost. Head-to-head method claims should be limited to studies that share these conditions; otherwise the comparison should be framed as evidence from different operating regimes [10, 16, 22].

4.2 Automated evaluators and human validation

Automated judges make repeated RAG experiments affordable, but their usefulness depends on agreement with human judgments about the intended construct. Rules, learned scorers, and LLM judges each embed assumptions about language, claim length, reference availability, and acceptable variation [16, 7]. Whereas automation provides scale and repeatability, human assessment remains necessary for detecting unfamiliar or domain-specific failures that were absent from an evaluator’s calibration data, even though it is slower and can also be inconsistent [6, 4].

Multi-dimensional evaluators offer better diagnosis than a single aggregate. ARES separates context relevance, faithfulness, and answer relevance and uses limited human annotations to calibrate evaluation across several tasks [16]. RAGVUE likewise seeks to identify whether problems arise from retrieval, reasoning, or grounding [12]. Neither decomposition removes the need to define labels carefully or to test judge agreement on the target distribution.

Cross-dataset attribution results quantify that calibration risk. Ding et al. The cited performance claim remains setup-bound in benchmark tasks and should not be generalized across automated evaluators and human validation without matching success rate and budget and cost model assumptions [7]. A prompt-based LLM judge avoided the chance-level LFQA collapse but was not uniformly best, was non-deterministic, and cost roughly one hundred times more [7]. Replacing a fixed scorer with an LLM judge therefore exchanges one error-and-cost profile for another.

Production evidence shows why a human-reviewed failure taxonomy matters. Cross-study interpretation in automated evaluators and human validation remains conditional on comparable benchmark tasks, success rate, and budget and cost model assumptions [6]. An industry review reports that RAG applications are commonly prototype-stage and domain-specific, that evaluation is predominantly human, and that security and data quality receive more attention than ethics, bias, and scalability [4]. Automated validation should be designed around such observed failures instead of assuming that generic faithfulness is exhaustive.

Human evaluation also needs an explicit protocol. Reports should describe annotator expertise, sampling, blinding, adjudication, agreement, and the severity assigned to different errors. For high-risk settings, oversampling rare but harmful cases can be more informative than estimating only average quality [6, 4]. These data can then support calibrated thresholds and reveal which automated dimensions are safe to use for routine monitoring. The practical design is a layered evaluation: deterministic checks for inexpensive invariants, learned or LLM judges for scalable triage, and human review for calibration, ambiguous cases, and high-severity errors. Each layer should be versioned and periodically revalidated after model, corpus, prompt, or domain changes [16, 12, 7].

5 Robustness, Transfer & Operational Validity

Benchmark validity is necessary but insufficient when the operating environment changes. Query form, corpus composition, feedback, domain, latency, and governance constraints can all move a RAG system outside the conditions under which its evaluation conclusion was obtained [13, 21, 1]. This chapter distinguishes controlled sensitivity tests from broader deployment evidence. The first reveals how performance varies under component and input shifts; the second asks whether evaluators and safeguards retain their meaning in new domains, especially where contradictory evidence or provenance errors carry high costs [23, 5, 8].

5.1 Sensitivity, adaptation, and query or corpus shift

Average performance on a fixed test set does not reveal whether a RAG system is stable. Whereas a static score summarizes one sample, robustness describes the distribution of outcomes after plausible changes to queries, documents, retrieval settings, or feedback under a fixed resource budget [13, 23, 1]. Evaluation should report both the mean and the variance, along with the frequency and severity of newly introduced failures.

Stage-resolved measurement is especially important under shift. Muthyam et al. note that final-answer accuracy can hide failures originating in preprocessing, retrieval, context packing, or generation [13]. A perturbation study can localize those failures by holding most of the pipeline constant while changing one component, such as chunk size, top-k, reranking, or context ordering. Without that localization, an apparently robust final score may reflect compensation between deteriorating retrieval and forgiving generation.

Query tests should represent realistic changes rather than only synthetic paraphrases. The TREC RAG challenge’s long narrative queries require reasoning over multi-sentence information needs, while imperfect visual queries introduce geometric distortion, degradation, and ambiguity [21, 23]. Medical benchmarks add specialized terminology and evidence requirements that generic QA collections may not represent [10]. Results should be stratified by perturbation type because resistance to wording changes does not imply resistance to modality or domain shift.

Adaptation adds a temporal dimension. Bang et al. argue that overall accuracy does not capture how systems respond after feedback is introduced [1]. Dynamic and parametric RAG approaches further make the use of external knowledge or stored parameters conditional, so evaluation must record when retrieval is invoked and how the policy changes over time [19]. Useful measures include pre/post-feedback performance, adaptation lag, regressions on earlier cases, retrieval frequency, and the cumulative cost of updating the system.

Corpus shift should include additions, deletions, stale material, and contradictions. A newly relevant document can improve recall while also displacing another source from a fixed context budget; outdated or conflicting material can remain highly similar to the query [8]. Snapshot dates and update procedures are therefore part of the experimental condition, not clerical metadata. A compact robustness matrix crosses query perturbation, corpus change, component setting, and feedback state, then records stage-level metrics, answer quality, latency, and cost. The available evidence is limited by abstract-level reporting, so the predictive value of individual perturbations

remains unclear. The unresolved empirical question is which small set of perturbations predicts deployed failures well enough to be run continuously rather than only during benchmark publication [13, 1, 23].

5.2 Domain transfer, trustworthiness, and deployment

Operational validity asks whether an evaluation conclusion still applies after the task, domain, evidence supply, or institutional constraints change. Accuracy and efficiency remain important, but trustworthiness has received less systematic attention in the RAG literature [24]. A deployment claim should additionally address provenance, contradiction handling, privacy, latency, maintenance, and the consequences of an incorrect answer.

Cross-dataset benchmarking is a starting point rather than proof of transfer. Cahoon et al. compare RAG methods on four datasets and report that TREX handles multiple open-domain question-answering types while the exercise also exposes limitations in current evaluation methods [5]. Whereas breadth can reveal inconsistent behavior, a result from a benchmark mixture cannot establish reliability for a new population unless the evaluator and failure taxonomy are validated there.

Domain structure changes what retrieval quality means. Flat Naive RAG pipelines are reported to degrade on heterogeneous corpora and multi-step queries when they lack explicit workflows [9]. In medical RAG, Javadi et al. find that highly similar abstracts can contradict one another and reduce factual accuracy, making similarity alone inadequate for reliable retrieval [8]. Contradiction-aware filtering and temporal provenance should therefore be evaluated alongside relevance in domains where evidence changes over time.

The healthcare evidence places semantic consistency and evidence age inside the trustworthiness target. Similarity-based retrieval should be supplemented by contradiction detection, source-date checks, and domain adjudication when old and new evidence may conflict [8]. The appropriate acceptance threshold should reflect harm severity rather than simply matching a generic QA benchmark [24].

Resource and governance constraints can alter the preferred system. A method that improves answer quality at high latency or maintenance cost may be unsuitable for interactive support, while an inexpensive evaluation may still omit important trustworthiness risks [5, 24]. Evaluation reports should state the hardware or service budget, model and index update policy, data-access rules, escalation path, and monitoring cadence so that readers can distinguish benchmark capability from deployable reliability.

Evidence for transfer is strongest when the same construct is revalidated in the target domain and followed after deployment. A useful sequence is external validation, domain-specific error discovery, calibrated human review, staged release, and ongoing drift measurement. Future work should test whether this sequence catches contradictions and provenance failures earlier than conventional relevance-and-faithfulness dashboards, and at what operational cost [5, 8, 24].

6 Discussion

The reviewed evidence rejects the idea that RAG evaluation can be reduced to one end-to-end score. Retrieval rank, evidence coverage, claim support, attribution, and task utility answer different questions, and their relationships vary with the dataset and system configuration [20, 7, 6]. A metric portfolio is therefore not redundant measurement; it is the mechanism by which compensating failures become visible. Three validity layers follow from the comparison. Target validity asks whether the selected measures correspond to the failure under study. Measurement validity asks whether datasets, references, metrics, and judges estimate those targets reliably. Operational validity asks whether the conclusion survives shifts in queries, corpora, domain, cost, and governance [16, 23, 13, 8]. Weakness at any layer limits the claim even when the other two are strong.

The quantitative examples show why portability cannot be presumed. Attribution evaluators reverse rank across datasets, an NLI scorer falls from 0.90 to 0.53 AUROC between short-claim and long-form settings, and selecting by performance elsewhere produces 0.172 mean held-out regret [7]. In deployment, deceptive grounding affects 7.8% of 740 drug-disease pairs and rises to 13.6% for recent drugs, despite failures being grounded in real documents [6]. These results concern different constructs, but both demonstrate that familiar aggregate checks can fail under a changed population or failure definition.

A practical evaluation program should begin with a protocol card, record stage-specific observables, and add robustness strata for plausible operating shifts. Automated judges can handle routine volume after target-domain calibration, while human review should concentrate on disagreement, novel cases, and high-severity risks [12, 4]. Costs, model versions, corpus snapshots, and update policies belong beside quality metrics because they determine whether the evaluation can be reproduced and maintained.

The evidence base limits how strongly these recommendations can be stated. The 48-paper core is broad enough to expose recurring validity problems, but much of the analysis relies on abstracts and does not verify every experimental detail, annotation protocol, or statistical procedure. The resulting framework is a synthesis of reported evidence and a reporting proposal, not proof that one metric portfolio is optimal for all RAG applications.

7 Conclusion

RAG systems should be evaluated as pipelines operating under explicit conditions, not as isolated answer generators. A defensible evaluation names the target construct, measures failures at the retrieval and generation stages, validates evaluators against the intended population, and tests whether conclusions survive realistic shifts [13, 16, 7]. This structure turns a leaderboard score into evidence that can guide diagnosis and deployment. The minimum reporting contract is concrete: query and corpus scope; chunking, top-k, and reranking settings; retrieval, coverage, faithfulness, attribution, and utility measures; human-calibration methods; per-stratum uncertainty; latency and cost; model and data versions; and post-deployment monitoring. High-risk domains also require entity- and contradiction-aware checks because real citations and high similarity do not guarantee correct grounding [6, 8].

No evaluator is universally reliable on the evidence reviewed here. The appropriate goal is a traceable chain from the decision a system owner must make, through the failure that would invalidate that decision, to the benchmark and judge capable of observing it. Future research can strengthen this chain by publishing target-domain validation, shift-sensitive results, and operational costs together rather than treating them as separate appendices [5, 24, 4].

A Tables

Table 1: Cross-section comparison map for the survey areas.

Survey area	Comparison focus	Representative evidence	Key refs
3.1 Retrieval quality and evidence coverage	retrieval recall and rank quality, evidence sufficiency and context coverage	Retrieval diagnostic studies vs Retrieval utility and coverage on retrieval recall and rank quality	[20, 9, 11, 7]
3.2 Generation faithfulness and source attribution	claim-level support and answer faithfulness, source-attribution correctness	Attribution and grounding vs Automated faithfulness evaluators on claim-level support and answer faithfulness	[14, 6, 16, 12]
4.1 Dataset, task, and metric design	corpus and query provenance, task coverage and difficulty	Domain and stress-test benchmarks vs Memory and retrieval augmentation on corpus and query provenance	[23, 10, 7, 16]
4.2 Automated evaluators and human validation	rule-based versus learned versus LLM-based judging, agreement with human assessment	Automated and model-based evaluators vs Human and deployment-grounded on rule-based versus learned versus LLM-based judging	[16, 12, 6, 7]
5.1 Sensitivity, adaptation, and query or corpus shift	query and corpus perturbation, retriever, chunker, and reranker sensitivity	Mapped subset A vs Mapped subset B on query and corpus perturbation	[13, 23, 10, 1]
5.2 Domain transfer, trustworthiness, and deployment	cross-domain metric and evaluator transfer, fairness and contradiction handling	Domain and deployment studies vs Memory and retrieval augmentation on cross-domain metric and evaluator transfer	[5, 9, 8, 3]

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